

Deggendorf Institute of Technology
Campus Cham
Course Name



MASTER THESIS

Sensor Fusion of Camera and Pressure Sensor Data for Cabin Monitoring System

submitted in the requirements for the degree of
Master of Engineering

by

CHETHAN SRINIVASAREDDY

Matriculation Number:

Supervisor: [SUPERVISOR NAME]

Cham, 2026

Acknowledgements

I would like to express my sincere gratitude to my supervisor, [SUPERVISOR NAME], for the continuous guidance, constructive feedback, and support provided throughout the course of this thesis. I would further like to thank the research group at Deggendorf Institute of Technology, Campus Cham, for providing access to the laboratory infrastructure used for pressure sensor and camera data collection. Finally, I am grateful to my family and friends for their patience and encouragement during this work.

Statement of Authorship

I hereby declare that I have authored this thesis independently, that I have not used other than the declared sources and aids, and that I have explicitly marked all material which has been quoted either literally or by content from the sources used. This thesis has not been submitted, either in part or as a whole, for any other academic degree or examination at this or any other institution. I am aware that any false statement regarding this declaration may lead to the invalidation of this thesis and may have legal consequences.

Cham, 2026

[CHETHAN SRINIVASAREDDY]

Abstract

In-cabin occupant monitoring is increasingly required by vehicle safety regulation and by advanced restraint systems that must adapt to who is seated where. This thesis investigates the fusion of RGB camera images with pressure heatmaps obtained from seat-embedded pressure sensors in order to jointly estimate seat occupancy, passenger class, occupant weight, and body pose. A synthetic pressure heatmap generation pipeline based on human body simulation is proposed to complement limited real sensor data, together with a Sim2Real validation procedure that quantifies the remaining domain gap. A multi-task deep learning fusion architecture combining camera and pressure branches is designed, trained, and evaluated against single-modality baselines through ablation studies. The expected contribution is a validated fusion pipeline and dataset methodology that improves robustness of occupant state estimation over either sensing modality used in isolation.

Keywords: cabin monitoring, sensor fusion, pressure heatmap, occupant classification, pose estimation, Sim2Real, multi-task learning

Contents

Acknowledgements	i
Statement of Authorship	ii
Abstract	iii
List of Abbreviations	ix
1 Introduction	1
1.1 background	1
1.2 Motivation	1
1.3 Problem Statement	2
1.4 Research Questions	3
1.5 Scope	4
1.6 Thesis Structure	4
2 Background and Fundamentals	5
2.1 Cabin Monitoring Systems	5
2.2 Pressure Sensing Technology	5
2.2.1 Working Principle and Sensor Types	5
2.2.2 Pressure Heatmap Representation	5
2.3 Camera Systems for In-Cabin Use	5
2.3.1 RGB and IR Cameras	5
2.3.2 Depth Cameras	6
2.4 Sensor Fusion Fundamentals	6
2.4.1 Fusion Levels	6
2.4.2 Deep Learning-Based Fusion	6
2.5 Multi-Task Learning	6
2.6 Human Body Modelling for Simulation	6
3 Related Work and State of the Art	7
3.1 Pressure-Based Occupant Detection and Classification	7
3.2 Camera-Based Cabin Monitoring	7
3.3 Multi-Modal Sensor Fusion Approaches	7
3.4 Synthetic Data Generation and Sim2Real Transfer	7
3.5 Gap Analysis and Positioning of This Work	8
4 System Requirements and Design	9
4.1 Use Case Definition and Target Scenarios	9
4.2 Functional Requirements	9
4.3 Non-Functional Requirements	9

4.4	System Architecture Overview	9
4.5	Design Decisions and Justification	10
5	Dataset Acquisition and Synthetic Data Generation	11
5.1	Ethics and Data Privacy Considerations	11
5.2	Synthetic Pressure Sensor Data Generation	11
5.2.1	Sensor Hardware and Seat Setup Specifications	11
5.2.2	Subjects and Experimental Protocol	11
5.2.3	Data Statistics and Limitations of setup	11
5.3	Camera Data Collection	11
5.3.1	Camera Hardware and Mounting	11
5.3.2	Recording Protocol	11
5.4	Synthetic Pressure Heatmap Generation	11
5.4.1	Human Body Simulation Model	11
5.4.2	Simulation Environment	12
5.4.3	Heatmap Rendering	12
5.4.4	Augmentation Strategy	12
5.5	Sim2Real Validation	12
5.6	Final data set pipeline	12
6	Data Preprocessing and Synchronisation	13
6.1	Pressure Heatmap Preprocessing	13
6.1.1	Noise Filtering and Normalisation	13
6.1.2	Spatial Alignment and Resizing	13
6.2	Camera Frame Preprocessing	13
6.2.1	Image Normalisation and Augmentation	13
6.2.2	Background Removal and ROI Extraction	13
6.3	Temporal Synchronisation	13
6.3.1	Timestamp Alignment Strategy	13
6.3.2	Handling Frame Rate Mismatch	14
6.3.3	Validation of Synchronisation Quality	14
6.4	Final Data Pipeline Overview	14
7	Sensor Fusion Architecture and Training Pipeline	15
7.1	Overall Architecture Design	15
7.2	Camera Branch	15
7.3	Pressure Heatmap Branch	15
7.4	Fusion Module	15
7.4.1	Feature-Level Fusion Strategy	15
7.4.2	Fusion Alternatives Considered and Rejected	15
7.5	Multi-Task Output Heads	16
7.5.1	Occupancy Detection Head	16
7.5.2	Passenger Classification Head	16

7.5.3	Weight Estimation Head	16
7.5.4	Pose Estimation Head	16
7.6	Loss Function Design and Task Balancing	16
7.7	Training Configuration	16
7.8	Implementation Details	17
8	Results and Evaluation	18
8.1	Experimental Setup and Evaluation Protocol	18
8.2	Occupancy Detection Results	18
8.3	Passenger Classification Results	18
8.4	Weight Estimation Results	19
8.5	Pose Estimation Results	19
8.6	Ablation Study: Modality Contribution	20
8.7	Sim2Real Transfer Evaluation	20
8.8	Discussion of Results and Failure Cases	21
9	Conclusion and Future Work	22
9.1	Summary of Contributions	22
9.2	Answers to Research Questions	22
9.3	Limitations	22
9.4	Future Work	22
9.5	Closing Remarks	22
A	Statement of Ethics / Consent Form Template	27
B	Full Dataset Statistics Table	28
C	Additional Results Figures	29
D	Electronic Data Carrier Contents	30

List of Figures

4.1 Block diagram of the proposed camera-pressure sensor fusion system architecture, from raw sensor acquisition to multi-task output estimation. 10

6.1 Overview of the data preprocessing and temporal synchronisation pipeline, from raw sensor streams to synchronised training samples. 14

8.1 Confusion matrix for passenger classification using the fused camera-pressure model on the held-out test set. 19

8.2 Qualitative comparison of predicted and ground-truth occupant pose for representative seating configurations. 20

C.1 Additional qualitative results across representative test samples, supplementing Section 8.8. 29

List of Tables

3.1 Comparison of related work against the scope of this thesis. 8

5.1 Final dataset composition by split, showing real and synthetic sample counts and
class distribution. 12

8.1 Occupancy detection results on the held-out test set. 18

8.2 Weight estimation error on the held-out test set. 19

8.3 Ablation study comparing camera-only, pressure-only, and fused model variants
across all output tasks. 20

8.4 Sim2Real transfer evaluation: performance on real-only test data for models
trained on real-only versus mixed real+synthetic data. 21

B.1 Extended per-class dataset statistics by split. 28

List of Abbreviations

Abbreviation	Meaning
CNN	Convolutional Neural Network
DMS	Driver Monitoring System
EKF	Extended Kalman Filter
FOV	Field of View
GDPR	General Data Protection Regulation
IMU	Inertial Measurement Unit
IR	Infrared
MSE	Mean Squared Error
OMS	Occupant Monitoring System
RGB-D	Red-Green-Blue plus Depth (image with depth channel)
ROI	Region of Interest
Sim2Real	Simulation-to-Reality (transfer)
ToF	Time-of-Flight
UKF	Unscented Kalman Filter
WRMSE	Weighted Root Mean Squared Error

1 Introduction

1.1 background

Occupant monitoring systems are becoming an increasingly important component of vehicle safety as vehicle interiors, occupant behaviour, and the role of the driver evolve with higher levels of driving automation. Conventional occupant-safety systems were primarily developed around relatively well-defined seating configurations and nominal upright postures. In more highly automated vehicles, however, occupants may adopt reclined, rotated, or otherwise non-nominal seating postures that differ from these conventional assumptions [1]. Automation also enables a broader range of non-driving-related activities, which can alter occupant posture and, in conditionally automated vehicles, influence the driver's readiness to resume control [2]. Consequently, occupant monitoring is evolving from simple seat-occupancy detection toward a more detailed description of occupant presence, body characteristics, and seating posture.

This transition is also reflected in current safety regulations and consumer-assessment frameworks. In the United States, FMVSS No. 208 introduced advanced-airbag requirements intended to reduce airbag-induced injury to vulnerable occupants while maintaining frontal-crash protection [3]. For conditionally automated driving, UN Regulation No. 157 requires a Driver Availability Recognition System to determine whether the driver is present and available to resume control [4], while European regulations additionally require advanced driver-distraction warning functions [5]. Euro NCAP's 2026 assessment framework extends occupant monitoring beyond the driver by introducing dedicated Occupant Monitoring requirements, occupant-stature classification, automatic passenger-airbag disabling, and additional credit for systems capable of optimising restraint and airbag performance for different occupant physiques [6].

The relevance of occupant-state information is expected to increase further as automated-driving systems progress toward vehicles that can operate without conventional driver controls. In June 2026, UNECE adopted the first global regulatory framework for fully automated driving systems. Existing occupant-protection regulations, however, do not yet fully cover all seating configurations that future automated interiors may enable. NHTSA's occupant-protection rule for automated vehicles without conventional driving controls is limited to vehicles with conventional forward-facing seating, while rear-facing and "campfire" seating configurations remain areas requiring further investigation. A UNECE review similarly concluded that side-facing or rear-facing seats and torso recline angles greater than 25° would require substantial changes to existing occupant-protection regulations. These developments indicate that future vehicle-safety systems will require increasingly accurate information about who occupies a seat and how that occupant is positioned.

1.2 Motivation

The need for richer occupant-state information is primarily safety-driven because occupant posture and physical characteristics influence the interaction between the human body and restraint systems. Real-world crash analysis using GIDAS data showed that belted reclined

occupants had higher odds of MAIS 2+, MAIS 3+, and MAIS 4+ injury than occupants seated upright [7]. In conditionally automated driving, non-driving postures can additionally increase takeover time and reduce takeover quality [2]. Reliable estimation of seated posture is therefore relevant both before a crash, where it can indicate driver readiness, and during a crash, where occupant geometry directly affects restraint interaction.

Occupant size or weight provides another relevant input because the appropriate restraint response is not identical for all occupants. Advanced-airbag systems can use information about occupant weight, size, and seating position to suppress deployment or adapt deployment intensity when necessary [8]. NHTSA notes that, when occupant weight or size is available, different inflation levels may be selected for smaller and larger adults [8]. Multi-stage airbag systems support such adaptation by independently controlling inflation stages to better match restraint force to occupant characteristics, as described in NHTSA guidance on advanced-airbag functionality [8]. This is particularly important because inappropriate or overly aggressive airbag deployment has historically caused serious and fatal injuries, especially for small occupants and occupants positioned close to the airbag module [9, 10]. Since an airbag must also be replaced after deployment [9], suppressing deployment when it is not required can provide an additional economic benefit.

Obtaining occupancy, posture, and occupant-size information reliably from a single sensing modality remains difficult. Camera-based monitoring provides rich semantic and geometric information, including body appearance, body landmarks, and upper-body posture, but its performance can deteriorate because of poor illumination, glare, shadows, or occlusion [11]. Seat-pressure sensing is independent of visible-light conditions and directly measures body-seat contact and load distribution, making it suitable for occupant detection and classification [12]. Its limitation is different: pressure measurements contain comparatively little semantic information, and similar contact distributions can be produced by different occupants or postures.

The two sensing modalities therefore provide complementary information rather than redundant measurements. RGB images primarily describe the visible geometry and semantic state of the occupant, whereas seat-pressure measurements describe the physical interaction between the occupant and the seat. Multimodal fusion offers a means of combining these complementary observations and can potentially provide more reliable predictions when one modality becomes uncertain or degraded [13]. Based on this premise, this thesis investigates whether feature-level fusion of monocular RGB cabin images and seat-pressure distribution maps improves the classification of seat occupancy, seated posture, and occupant size or weight class compared with equivalent RGB-only and pressure-only models.

1.3 Problem Statement

Reliable occupant monitoring requires the estimation of multiple occupant states, including seat occupancy, occupant category, posture, and occupant weight class. This requirement is becoming increasingly relevant as vehicle-safety assessment frameworks such as Euro NCAP place greater emphasis on occupant monitoring and the adaptation of safety functions to occupant characteristics [6]. Reliably estimating these states from a single sensing modality, however, remains challenging.

Camera-based systems provide rich semantic and geometric information for detecting occupants and estimating body posture, but their performance can deteriorate under poor illumination, glare, shadows, and partial occlusion [11]. Seat-pressure sensing avoids these visual limitations and can capture body-seat contact, load distribution, and posture-related pressure patterns. Even relatively low-resolution pressure configurations have demonstrated useful posture-recognition capability [14]. Nevertheless, recent pressure-based studies show that performance can degrade when seating conditions change and that geometrically similar postures may produce ambiguous pressure patterns [15]. Thus, the two sensing modalities provide complementary occupant information while exhibiting different failure modes.

Multimodal in-cabin monitoring has already demonstrated that heterogeneous signals such as video, seat pressure, and physiological measurements can be combined successfully for specific tasks such as driver takeover-state recognition [16–18]. However, existing studies provide limited evidence on whether combining camera images and seat-pressure maps improves a unified set of occupant-monitoring tasks—particularly occupancy, occupant classification, posture, and occupant mass classification—relative to equivalent unimodal baselines.

The central problem addressed in this thesis is therefore to determine whether feature-level fusion of monocular RGB and seat-pressure information provides a measurable improvement in occupant-state estimation compared with camera-only and pressure-only models. The comparison must be conducted using synchronized data, equivalent task definitions, and consistent evaluation criteria so that any performance gain can be attributed to multimodal fusion rather than differences in data or evaluation conditions.

1.4 Research Questions

"To what extent does feature-level fusion of synchronized RGB cabin images and low-resolution seat-pressure distribution maps improve the classification of seat occupancy, seated posture, and occupant size class compared with equivalent RGB-only and pressure-only models."

1. RQ1.1: How do the RGB-only and pressure-only models compare in classification performance across seat occupancy, seated posture, and occupant weight classification?
2. RQ1.2: To what extent does the fused RGB–pressure model improve classification performance over the best-performing unimodal baseline for each classification task?
3. RQ1.3: What contribution do sensor weighted combination(WC) and cross-sensor correlation(CS) fusion framework does to the classification performance of the RGB–pressure fusion model?
4. RQ1.4: How robust is RGB–pressure fusion to controlled degradation of either sensing modality compared with the corresponding unimodal models?

Terminology: "modal/modality" refers to a distinct source of sensory information. RGB images constitute the visual modality, seat-pressure distribution maps constitute the pressure modality. "model" refers to the learned computational architecture used to process one or both modalities.

Occupant state = the overall concept being estimated. Occupant-state attributes = occupancy, posture, and size/weight class. Classes = the discrete labels within each attribute.

1.5 Scope

This thesis investigates supervised, per-seat occupant-state classification using two heterogeneous sensing modalities: monocular RGB cabin images and low-resolution seat-pressure distribution maps. The study is restricted to adult occupants and considers three target attributes: seat occupancy, seated posture, and occupant mass class. RGB-only and pressure-only models are evaluated as unimodal baselines and compared with a feature-level RGB–pressure fusion model under the same dataset partitions and evaluation criteria.

Seat occupancy is formulated using the classes *occupied* and *empty*. For occupied seats, posture is classified as *upright*, *forward lean*, *reclined*, or *side lean*. Occupant mass is represented using three study-specific classes: *light* (49–< 63.35,kg), *medium* (63.35–< 89.45,kg), and *heavy* (89.45–101.2,kg). The detailed geometric definitions of the posture classes and the justification of the mass-class boundaries are presented in the experimental methodology.

The pressure modality is represented using synthetic low-resolution pressure maps generated to reproduce the spatial configuration of the target pressure-sensor system. Consequently, the conclusions regarding pressure-based and multimodal performance are limited to this simulated representation. Transfer to a physical pressure mat and quantitative validation of the simulated pressure distributions against real measurements are outside the scope of this study.

Children, occupants outside the defined adult mass range, postures outside the predefined classes, production-vehicle deployment, direct restraint-system control, and validation under crash conditions are not considered. The objective is limited to evaluating whether RGB–pressure fusion improves the classification performance and robustness of the defined occupant-state tasks relative to the corresponding unimodal models.

1.6 Thesis Structure

2 Background and Fundamentals

2.1 Cabin Monitoring Systems

Cabin monitoring systems are commonly divided into driver monitoring systems (DMS), which focus on the driver's attentiveness and drowsiness, and occupant monitoring systems (OMS), which assess the presence, position, and classification of all vehicle occupants. Regulatory bodies such as Euro NCAP have progressively increased the weight given to occupant status monitoring in their safety rating protocols, driving adoption of both camera-based and pressure-based sensing across vehicle segments [19]. Typical OMS use cases include seatbelt reminder logic, smart airbag suppression for child seats, and post-crash occupant status reporting for emergency services.

2.2 Pressure Sensing Technology

2.2.1 Working Principle and Sensor Types

Seat-integrated pressure sensors typically use resistive, capacitive, or piezoelectric transduction principles arranged in a grid to produce a spatially resolved pressure map of the seat surface. Resistive force sensing resistor (FSR) arrays are low-cost and widely used in automotive seat occupancy classification, while capacitive mats offer higher sensitivity at increased manufacturing cost [20]. The choice of sensor technology directly affects spatial resolution, sampling rate, and susceptibility to drift over the vehicle's operating temperature range.

[21] This is paper on lower resolution pressure sensor array for driver posture monitoring and pose estimation.

2.2.2 Pressure Heatmap Representation

The raw output of a pressure sensor grid is commonly represented as a two-dimensional heatmap, where each pixel encodes the normalised pressure value at a corresponding physical location on the seat surface. This representation allows pressure data to be processed with the same convolutional architectures typically applied to single-channel images, which is a key enabler for the fusion architecture proposed in Chapter 7.

2.3 Camera Systems for In-Cabin Use

2.3.1 RGB and IR Cameras

Standard RGB cameras provide high-resolution colour imagery suitable for detailed pose and appearance analysis but perform poorly under low-light or backlit conditions common in vehicle cabins. Near-infrared (IR) cameras paired with active IR illumination maintain consistent image quality irrespective of ambient lighting and are therefore preferred in production driver and occupant monitoring systems [22].

2.3.2 Depth Cameras

2.4 Sensor Fusion Fundamentals

2.4.1 Fusion Levels

Sensor fusion approaches are typically categorised into data-level fusion (combining raw signals before any feature extraction), feature-level fusion (combining learned intermediate representations), and decision-level fusion (combining independent per-modality predictions). Feature-level fusion generally offers the best balance between representational flexibility and robustness to modality-specific noise, and is the strategy adopted for the architecture in this thesis [23].

2.4.2 Deep Learning-Based Fusion

Deep learning-based fusion replaces hand-crafted process and measurement models with learned convolutional or attention-based feature extractors per modality, followed by a learned fusion module, building on the general deep learning foundations of representation learning and gradient-based training described in [24]. This approach scales naturally to high-dimensional, spatially structured inputs such as camera images and pressure heatmaps, and can be trained end-to-end jointly with the downstream task heads [25].

2.5 Multi-Task Learning

Multi-task learning trains a single shared backbone with multiple task-specific output heads, allowing related tasks to regularise one another and share statistical strength, which is particularly valuable when labelled data for some tasks (e.g. weight estimation) is scarcer than for others (e.g. binary occupancy). A central challenge in multi-task learning is balancing the loss contributions of tasks with different scales and convergence rates, which is addressed explicitly for this thesis in Section 7.6.

2.6 Human Body Modelling for Simulation

Parametric human body models represent body shape and pose using a low-dimensional set of shape and joint-angle parameters, enabling generation of anatomically plausible body meshes across a wide population of statures and weights. Combined with a physics-based seat contact simulation, such models allow synthetic pressure heatmaps to be rendered for arbitrary occupant configurations without requiring a physical subject, which forms the basis of the synthetic data generation pipeline described in Section 5.4 [26].

3 Related Work and State of the Art

3.1 Pressure-Based Occupant Detection and Classification

Early automotive seat occupancy detection relied on simple threshold-based binary weight switches, which were later extended to spatially resolved pressure mats capable of distinguishing occupant classes beyond a simple occupied/unoccupied signal [20]. Subsequent work applied regression neural networks directly to pressure heatmaps to estimate continuous occupant weight, demonstrating that spatial pressure distribution carries substantially more information than aggregate load alone [27]. A recent survey consolidates these pressure-sensing approaches and highlights classification accuracy, sensor cost, and robustness to seat cover variation as the primary axes along which such systems are compared [28].

3.2 Camera-Based Cabin Monitoring

Camera-based approaches to cabin monitoring have progressed from frontal-face driver drowsiness detection toward full-cabin, multi-occupant pose estimation. Infrared camera systems combined with real-time pose estimation pipelines have been shown to operate reliably across day and night conditions inside the vehicle cabin [22]. Lightweight convolutional architectures tailored to the automotive embedded compute budget have been proposed specifically for occupant pose classification [29], while RGB-D fusion methods combine colour and depth cues to improve robustness against variable illumination and occluding objects such as blankets or bags placed on seats [30].

3.3 Multi-Modal Sensor Fusion Approaches

Beyond the automotive cabin monitoring domain, multimodal deep fusion has been studied extensively for tasks such as autonomous driving perception and human activity recognition. A broad survey categorises fusion architectures by fusion level and highlights feature-level fusion with learned attention mechanisms as consistently outperforming naive data-level concatenation [23]. Cross-modal attention fusion between camera and depth sensors has further been shown to improve robustness specifically for human activity and pose recognition tasks [25], providing direct architectural inspiration for the camera-pressure fusion module proposed in this thesis.

3.4 Synthetic Data Generation and Sim2Real Transfer

Given the cost and privacy constraints of collecting large real pressure sensor datasets across diverse occupant populations, synthetic data generation using simulated human bodies and physics-based seat contact models has emerged as a practical complement to real data collection [26]. Sim2Real studies quantify the remaining statistical gap between simulated and real sensor readings and propose domain adaptation or fine-tuning strategies to close it; a recent study specifically targeting seat pressure heatmaps reports meaningful performance retention

when models trained on a mixture of synthetic and real data are evaluated on real-only test sets [31].

?? provides a comprehensive review of human body simulation techniques for generating synthetic pressure heatmaps, including parametric body models, pose libraries, and physics-based contact simulations.

3.5 Gap Analysis and Positioning of This Work

While the works reviewed above individually address pressure-based classification, camera-based pose estimation, generic multimodal fusion, or synthetic pressure data generation, none jointly combine camera and pressure heatmap fusion with a synthetic data generation and Sim2Real validation pipeline for the specific multi-task set of occupancy, passenger classification, weight, and pose estimation. Table 3.1 summarises this positioning relative to the reviewed literature.

Table 3.1: Comparison of related work against the scope of this thesis.

Work	Modalities	Outputs Estimated	Synthetic Data	Real-Time
[20]	Pressure	Occupancy class	No	Yes
[32]	Pressure (sim)	Occupancy class	Yes	No
[27]	Pressure	Weight	No	No
[22]	Camera (IR)	Pose	No	Yes
[29]	Camera (RGB)	Pose class	No	Yes
[30]	Camera (RGB-D)	Pose	No	Partial
[25]	Camera + Depth	Activity class	No	Partial
[26]	Pressure (sim)	Occupancy class	Yes	No
[31]	Pressure	Statistical validation	Yes	No
This thesis	Camera + Pressure	Occupancy, class, weight, pose	Yes	Target

4 System Requirements and Design

4.1 Use Case Definition and Target Scenarios

The target system is a passenger car cabin monitoring function that continuously estimates, for each seating position, whether the seat is occupied, which passenger class occupies it (e.g. adult, or unknown), the occupant weight class, and the occupant's body pose. Target scenarios include front and rear seating positions, a range of occupant statures and clothing.

4.2 Functional Requirements

The system shall satisfy the following functional requirements:

- FR-01** The system shall detect, per seating position, whether the seat is occupied or unoccupied.
- FR-02** The system shall classify the seat occupant into predefined passenger classes (e.g. adult, or unknown).
- FR-03** The system shall estimate the occupant's weight within a defined tolerance for adult.
- FR-04** The system shall estimate the occupant's body pose for defined set pose categories..

4.3 Non-Functional Requirements

The system shall additionally satisfy the following non-functional requirements:

- NFR-01** Occupancy detection accuracy shall exceed a defined target (e.g. 98%) across the validation dataset described in Chapter 5.
- NFR-02** The system shall remain robust to sensor noise, partial occlusion, and seat cover variation without requiring per-vehicle recalibration.
- NFR-03** All data collection, storage, and processing shall comply with the General Data Protection Regulation (GDPR) and applicable local data protection law.
- NFR-05** The trained model shall degrade gracefully if one modality (camera or pressure) becomes temporarily unavailable.

4.4 System Architecture Overview

The proposed system architecture consists of four main stages: sensor data acquisition (camera and pressure sensor array), preprocessing and temporal synchronisation, the multi-task fusion model, and the output interface that exposes per-seat occupancy, classification, weight, and pose estimates to downstream vehicle systems. Figure 4.1 illustrates this pipeline at block-diagram level, showing the flow from raw sensor signals through synchronised preprocessing into the shared fusion backbone and task-specific output heads described in detail in Chapter 7.

[33] This is a novel deep fusion architecture that can be used .

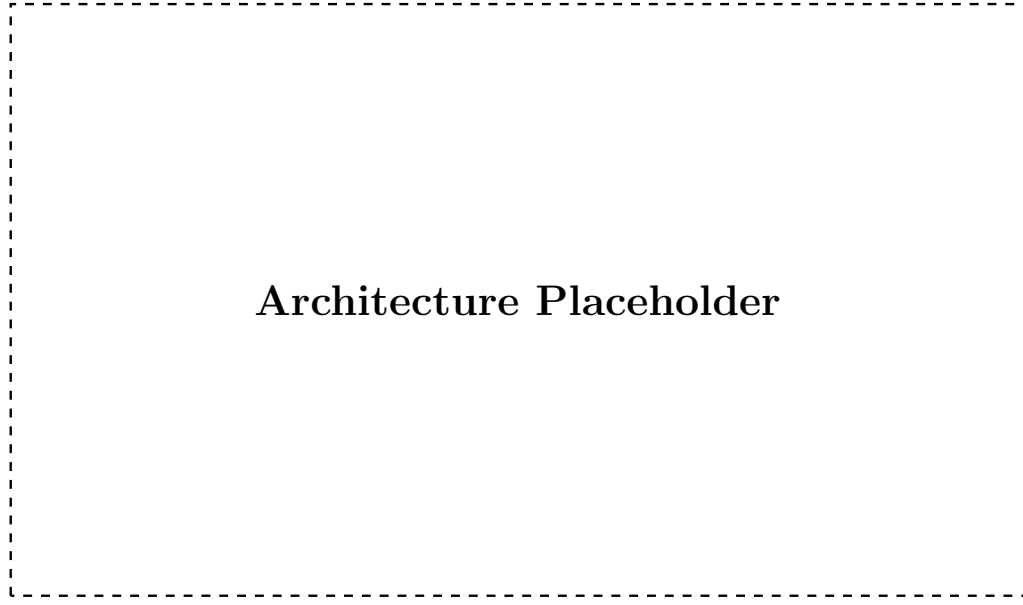


Figure 4.1: Block diagram of the proposed camera-pressure sensor fusion system architecture, from raw sensor acquisition to multi-task output estimation.

4.5 Design Decisions and Justification

Feature-level fusion was selected over data-level and decision-level fusion because it allows each modality-specific branch to learn representations suited to its own signal characteristics while still enabling the network to learn cross-modal correlations before the final task predictions are made, consistent with the fusion levels discussed in Section 2.4.1. A multi-task learning formulation was chosen over four independent single-task models to exploit shared structure between occupancy, classification, weight, and pose labels, and to reduce the total inference compute footprint relative to running four separate networks.

5 Dataset Acquisition and Synthetic Data Generation

5.1 Ethics and Data Privacy Considerations

5.2 Synthetic Pressure Sensor Data Generation

5.2.1 Sensor Hardware and Seat Setup Specifications

5.2.2 Subjects and Experimental Protocol

5.2.3 Data Statistics and Limitations of setup

The resulting real pressure dataset provides ground-truth labels for occupancy, passenger class, weight, and coarse pose for each recorded configuration, but is limited in overall subject diversity and recording duration due to the practical constraints of laboratory-based data collection. These limitations directly motivate the synthetic data generation pipeline described in Section 5.4.

5.3 Camera Data Collection

5.3.1 Camera Hardware and Mounting

Camera recordings were captured using an infrared-sensitive camera module mounted in the vehicle headliner, providing a field of view that covers the target seating positions under both daylight and night-time illumination, following mounting conventions similar to those reported in [22].

5.3.2 Recording Protocol

Camera frames were recorded continuously and time-stamped alongside the pressure sensor stream for every subject session described in Section 5.2.2, ensuring that each seating configuration has a corresponding synchronised camera-pressure sample pair available for the synchronisation pipeline in Chapter 6.

5.4 Synthetic Pressure Heatmap Generation

5.4.1 Human Body Simulation Model

Synthetic pressure heatmaps are generated by simulating a parametric human body model with sampled shape and pose parameters seated on a physics-based model of the seat geometry, following the simulation approach proposed by [26]. Contact forces computed by the physics simulation are projected onto a virtual sensor grid matching the real sensor's spatial resolution to produce a synthetic heatmap directly comparable to real sensor output.

5.4.2 Simulation Environment

5.4.3 Heatmap Rendering

The rendering pipeline samples occupant shape, pose, and seating configuration parameters, runs the physics-based contact simulation, and rasterises the resulting per-cell contact pressure into the same heatmap format used for real sensor data, allowing synthetic and real samples to be interchanged during model training without format-specific handling.

5.4.4 Augmentation Strategy

To increase the diversity of the synthetic dataset beyond the sampled body model parameters, additional augmentations are applied, including random sensor noise injection, simulated calibration drift, and small random seat-position offsets, so that models trained on synthetic data are not overly reliant on idealised, noise-free heatmaps.

5.5 Sim2Real Validation

5.6 Final data set pipeline

Table 5.1 s.

Table 5.1: Final dataset composition by split, showing real and synthetic sample counts and class distribution.

Split	Real Samples	Synthetic Samples	Total	Class Distribution
Training				
Validation				
Test				
Total				—

6 Data Preprocessing and Synchronisation

6.1 Pressure Heatmap Preprocessing

6.1.1 Noise Filtering and Normalisation

Raw pressure heatmaps are first passed through a spatial median filter to suppress isolated sensor cell dropouts and transient noise spikes, after which pixel values are normalised to a fixed range using the per-sensor calibration parameters obtained during the setup described in Section 5.2.1. This normalisation step ensures that synthetic and real heatmaps share a consistent value range prior to being combined in the mixed training dataset.

6.1.2 Spatial Alignment and Resizing

Because the physical sensor grid resolution differs from the target input resolution expected by the pressure branch described in Section 7.3, heatmaps are resampled onto a fixed target grid size using bilinear interpolation, and a fixed reference coordinate frame is applied so that seat regions (seat base, backrest) occupy consistent pixel regions across all recordings.

6.2 Camera Frame Preprocessing

6.2.1 Image Normalisation and Augmentation

Camera frames are resized to a fixed input resolution and normalised using per-channel mean and standard deviation statistics computed over the training split. During training, standard augmentations such as random brightness and contrast jitter, small rotations, and horizontal flipping (where anatomically valid for the seating position) are applied to improve robustness to the lighting and pose variation present in the target deployment scenarios described in Section 4.1.

6.2.2 Background Removal and ROI Extraction

A fixed region of interest (ROI) corresponding to each monitored seating position is cropped from the full camera frame using the known camera mounting geometry, and static background regions (e.g. door panels, window area) outside the seat envelope are masked out to reduce irrelevant visual variation presented to the camera branch.

6.3 Temporal Synchronisation

6.3.1 Timestamp Alignment Strategy

Camera frames and pressure heatmap samples are each recorded with hardware or system-clock timestamps, and are aligned by matching each pressure sample to the camera frame with the closest timestamp within a fixed maximum tolerance window, discarding samples that fall outside this window to avoid pairing temporally inconsistent data.

6.3.2 Handling Frame Rate Mismatch

Because the camera and pressure sensor operate at different native sampling rates, the higher-rate stream is downsampled to match the lower-rate stream at inference time, while during dataset construction all valid synchronised pairs are retained to maximise the number of training samples available from each recording session.

6.3.3 Validation of Synchronisation Quality

Synchronisation quality is validated by manually inspecting a sample of paired camera-pressure recordings for temporal consistency (e.g. confirming that a visible occupant movement in the camera frame corresponds to a matching pressure distribution change), and by measuring the distribution of timestamp offsets across all paired samples in the dataset.

6.4 Final Data Pipeline Overview

Figure 6.1 illustrates the complete preprocessing pipeline, from raw camera and pressure sensor streams through per-modality preprocessing and temporal synchronisation to the production of paired training samples consumed by the fusion architecture described in Chapter 7.

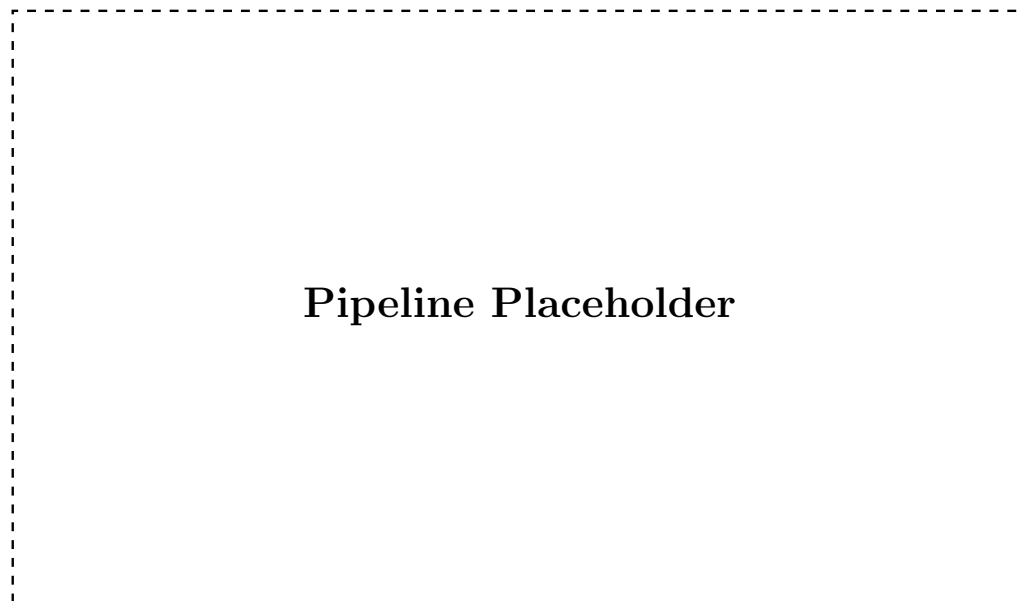


Figure 6.1: Overview of the data preprocessing and temporal synchronisation pipeline, from raw sensor streams to synchronised training samples.

7 Sensor Fusion Architecture and Training Pipeline

7.1 Overall Architecture Design

The proposed architecture follows a two-branch feature-level fusion design, in which a camera branch and a pressure heatmap branch each extract modality-specific feature representations that are combined by a fusion module before being passed to four task-specific output heads for occupancy, passenger classification, weight, and pose estimation. This design directly implements the feature-level fusion strategy motivated in Section 2.4.1 and the design decisions summarised in Section 4.5.

7.2 Camera Branch

The camera branch consists of a convolutional neural network backbone that processes the preprocessed, ROI-cropped camera frame described in Section 6.2.2 and produces a compact spatial feature embedding. A backbone pretrained on a large-scale image dataset is used as initialisation to compensate for the comparatively limited size of the in-cabin camera dataset, with the final classification layers removed and replaced by a projection to the shared fusion feature dimension.

7.3 Pressure Heatmap Branch

The pressure heatmap branch applies a separate, smaller convolutional network directly to the single-channel, spatially aligned pressure heatmap produced by the preprocessing pipeline in Section 6.1.2, producing a feature embedding of the same dimensionality as the camera branch output. A smaller network is used for this branch given the lower spatial resolution and information density of pressure heatmaps compared to camera images.

7.4 Fusion Module

7.4.1 Feature-Level Fusion Strategy

The camera and pressure feature embeddings are combined using a learned cross-modal attention fusion module, in which each modality’s features attend to the other before being concatenated and projected to a shared fused representation, following the cross-modal attention design demonstrated for camera-depth fusion in [25]. This fused representation is then passed to the multi-task output heads described in Section 7.5.

7.4.2 Fusion Alternatives Considered and Rejected

Simple feature concatenation followed by a fully connected layer was implemented as a baseline fusion strategy but was found to underweight the lower-dimensional pressure branch relative to the camera branch; decision-level fusion, in which independent per-modality predictions are averaged, was also considered but rejected because it cannot model cross-modal correlations

such as a camera-visible object combined with a pressure signature inconsistent with a human occupant. Both alternatives are retained as baselines for the ablation study reported in Section 8.6.

7.5 Multi-Task Output Heads

7.5.1 Occupancy Detection Head

The occupancy detection head is a lightweight fully connected layer followed by a sigmoid activation, producing a binary occupied/unoccupied probability for the corresponding seating position.

7.5.2 Passenger Classification Head

The passenger classification head is a fully connected layer followed by a softmax activation over the predefined passenger classes (e.g. adult, child, child seat, object), sharing the fused feature representation with the other task heads.

7.5.3 Weight Estimation Head

The weight estimation head is a small fully connected regression network that outputs a continuous estimate of occupant weight in kilograms, trained only on samples labelled as human occupants.

7.5.4 Pose Estimation Head

The pose estimation head outputs either a fixed set of body keypoint coordinates or a discrete pose category, depending on the granularity of pose annotation available in the dataset described in Chapter 5.

7.6 Loss Function Design and Task Balancing

Each task head is trained with a task-appropriate loss term: binary cross-entropy for occupancy, categorical cross-entropy for passenger classification, mean squared error for weight estimation, and a keypoint or classification loss for pose estimation. The combined multi-task training objective is a weighted sum of the individual task losses,

$$\mathcal{L}_{\text{total}} = \lambda_{\text{occ}} \mathcal{L}_{\text{occ}} + \lambda_{\text{cls}} \mathcal{L}_{\text{cls}} + \lambda_{\text{wt}} \mathcal{L}_{\text{wt}} + \lambda_{\text{pose}} \mathcal{L}_{\text{pose}}, \quad (7.1)$$

where λ_{occ} , λ_{cls} , λ_{wt} , and λ_{pose} are task weighting coefficients. These coefficients are tuned to balance the differing scales and convergence rates of the four loss terms, as discussed in the multi-task learning background in Section 2.5.

7.7 Training Configuration

The model is implemented in PyTorch and trained using the Adam optimiser with a cosine learning-rate decay schedule, an initial learning rate selected via a short learning-rate range test,

and a batch size chosen to fit the available GPU memory. Training is performed on a workstation equipped with a CUDA-capable GPU, with early stopping based on validation loss to reduce overfitting given the moderate dataset size described in Section 5.6.

7.8 Implementation Details

The camera and pressure branches, fusion module, and task heads are implemented as a single end-to-end trainable PyTorch model, with modality-specific data loaders that apply the preprocessing steps described in Chapter 6 on the fly during training. Model checkpoints corresponding to the best validation performance are retained for the evaluation reported in Chapter 8.

8 Results and Evaluation

8.1 Experimental Setup and Evaluation Protocol

All results reported in this chapter are computed on the held-out test split of the dataset described in Section 5.6, using the model checkpoint selected by best validation performance as described in Section 7.7. Occupancy and passenger classification are evaluated using accuracy and F1-score, weight estimation is evaluated using mean absolute error and weighted root mean squared error (WRMSE), and pose estimation is evaluated using mean per-keypoint error or classification accuracy depending on the final pose representation chosen in Section 7.5.4.

8.2 Occupancy Detection Results

The fused model is expected to achieve high occupancy detection accuracy given the strong, complementary occupancy signal present in both the camera and pressure modalities. Table 8.1 reports placeholder occupancy detection results across the evaluated models.

Table 8.1: Occupancy detection results on the held-out test set.

Model	Accuracy (%)	F1-score
Camera-only		
Pressure-only		
Fused (ours)		

8.3 Passenger Classification Results

Passenger classification is expected to benefit most from fusion in cases where an object placed on the seat produces a pressure signature ambiguous with a small child but is visually distinguishable from the camera view, or vice versa under partial visual occlusion. Figure 8.1 presents a placeholder confusion matrix for the fused model across all passenger classes.

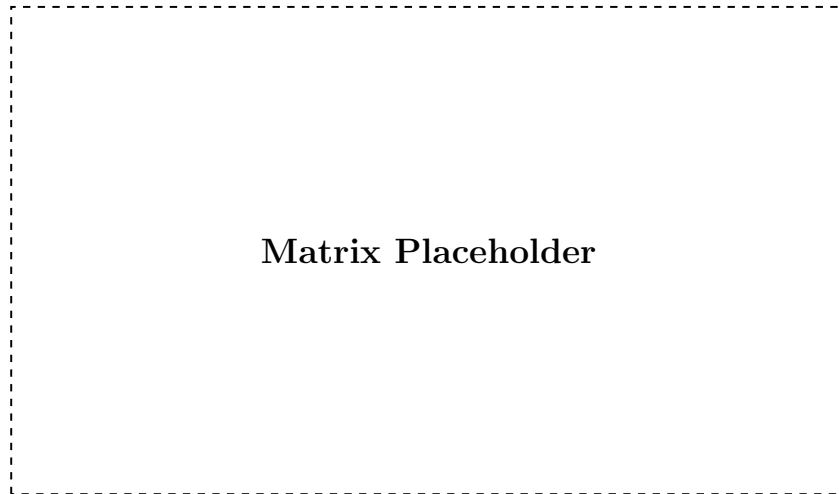


Figure 8.1: Confusion matrix for passenger classification using the fused camera-pressure model on the held-out test set.

8.4 Weight Estimation Results

Table 8.2 reports placeholder weight estimation error metrics, comparing the fused model against single-modality baselines.

Table 8.2: Weight estimation error on the held-out test set.

Model	MAE (kg)	WRMSE (kg)
Camera-only		
Pressure-only		
Fused (ours)		

8.5 Pose Estimation Results

Pose estimation performance is reported in terms of the metric selected in Section 7.5.4 (mean per-keypoint error or classification accuracy). Figure 8.2 shows placeholder qualitative examples comparing predicted and ground-truth pose across representative seating configurations.

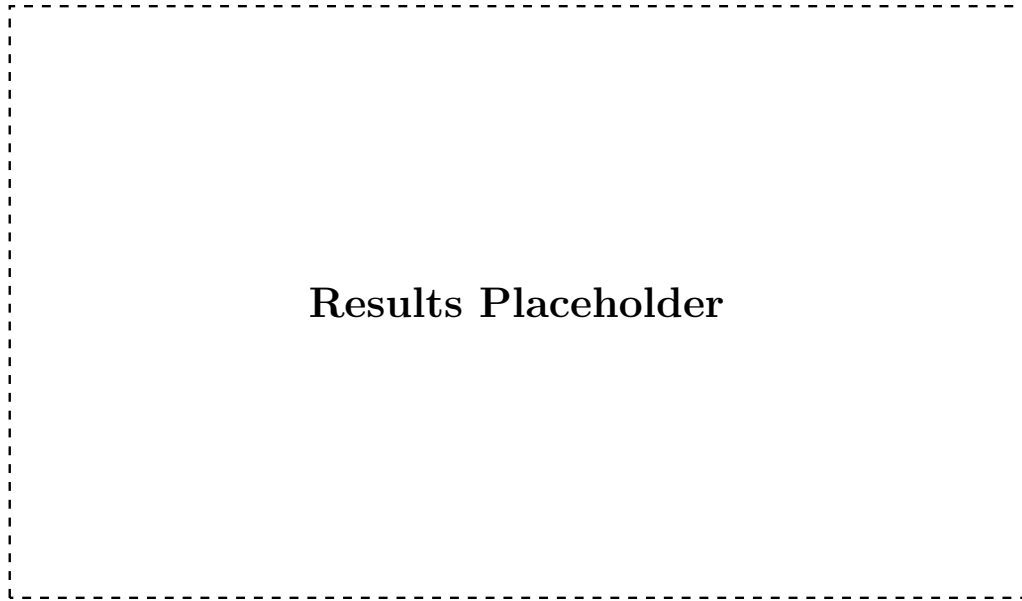


Figure 8.2: Qualitative comparison of predicted and ground-truth occupant pose for representative seating configurations.

8.6 Ablation Study: Modality Contribution

To quantify the individual contribution of each modality, the fused model is compared against camera-only and pressure-only variants trained with the same task heads and training configuration. Table 8.3 summarises this comparison across all four output tasks.

Table 8.3: Ablation study comparing camera-only, pressure-only, and fused model variants across all output tasks.

Metric	Camera-only	Pressure-only	Fused
Occupancy accuracy (%)			
Classification accuracy (%)			
Weight MAE (kg)			
Pose error			

8.7 Sim2Real Transfer Evaluation

To evaluate Sim2Real transfer, a model is trained on the mixed real and synthetic training set described in Section 5.6 and evaluated exclusively on the real-only subset of the test split. Table 8.4 compares this model against a variant trained only on real data, to isolate the effect of synthetic data augmentation on real-world performance.

Table 8.4: Sim2Real transfer evaluation: performance on real-only test data for models trained on real-only versus mixed real+synthetic data.

Metric	Trained on Real-only	Trained on Mixed (Real+Synthetic)
Occupancy accuracy (%)		
Classification accuracy (%)		
Weight MAE (kg)		

8.8 Discussion of Results and Failure Cases

Overall, the fused model is expected to outperform both single-modality baselines across most tasks, consistent with the complementary nature of camera and pressure information motivated in Section ???. Anticipated failure cases include ambiguous object-versus-child-seat pressure signatures under camera occlusion, and pose estimation errors for heavily reclined seating positions not well represented in the training data.

9 Conclusion and Future Work

9.1 Summary of Contributions

This thesis presented a multi-task deep learning architecture that fuses camera images and seat-embedded pressure heatmaps to jointly estimate occupancy, passenger class, weight, and pose for automotive cabin monitoring. Key contributions include a synthetic pressure heatmap generation pipeline based on human body simulation, a Sim2Real validation procedure quantifying the domain gap between real and synthetic heatmaps, a camera-pressure synchronisation pipeline, and an empirical comparison of fusion strategies through ablation studies.

9.2 Answers to Research Questions

Revisiting the research questions posed in Section 1.4: regarding RQ1, the ablation study in Section 8.6 indicates ; regarding RQ2, the Sim2Real validation in Section 5.5 shows ; regarding RQ3, the comparison of fusion alternatives in Section 7.4.2 demonstrates ; regarding RQ4, the Sim2Real transfer evaluation in Section 8.7 shows .

9.3 Limitations

The real pressure and camera dataset collected for this thesis is limited in subject diversity and recording duration due to practical laboratory constraints, and the synthetic data generation pipeline relies on simplifying assumptions in the human body and seat contact simulation that may not capture all real-world seating behaviours. Additionally, the evaluation was conducted offline on recorded data rather than on embedded target hardware, so real-time performance under production constraints remains to be confirmed.

9.4 Future Work

Future work could extend the real data collection campaign to a larger and more diverse subject population, incorporate additional sensing modalities such as depth or seatbelt tension sensors, and evaluate the proposed architecture on embedded automotive hardware to validate real-time feasibility under the non-functional requirements defined in Section 4.3. Further investigation into domain adaptation techniques could also help further close the Sim2Real gap identified in Section ??.

9.5 Closing Remarks

The fusion of camera and pressure sensor data investigated in this thesis represents a step toward more robust, privacy-conscious cabin monitoring systems capable of supporting the next generation of adaptive vehicle safety features. The methodology and findings presented here are intended to inform both future academic research and practical system design in this area.

Bibliography

- [1] Timothy L. McMurphy et al. "Crash Safety Concerns for Out-of-Position Occupant Postures: A Look Toward Safety in Highly Automated Vehicles". In: *Traffic Injury Prevention* 19.6 (2018), pp. 582–587. DOI: 10.1080/15389588.2018.1458306. URL: <https://www.tandfonline.com/doi/full/10.1080/15389588.2018.1458306>.
- [2] Mingming Zhao et al. "Effects of Non-Driving Related Postures on Takeover Performance During Conditionally Automated Driving". In: *Accident Analysis & Prevention* 208 (2024), p. 107793. ISSN: 0001-4575. DOI: 10.1016/j.aap.2024.107793. URL: <https://www.sciencedirect.com/science/article/pii/S0001457524003385>.
- [3] National Highway Traffic Safety Administration. *Federal Motor Vehicle Safety Standard No. 208: Occupant Crash Protection*. 49 CFR §571.208, U.S. Department of Transportation. Accessed: TODO. 2023.
- [4] United Nations Economic Commission for Europe. *UN Regulation on Automated Lane Keeping Systems Is Milestone for Safe Introduction of Automated Vehicles in Traffic*. 2020. URL: <https://unece.org/media/press/348314> (visited on 08/27/2026).
- [5] European Commission. *Commission Delegated Regulation (EU) 2023/2590 of 13 July 2023 Supplementing Regulation (EU) 2019/2144 Concerning Advanced Driver Distraction Warning Systems*. Official Journal of the European Union. 2023. URL: http://data.europa.eu/eli/reg_del/2023/2590/oj (visited on 08/27/2026).
- [6] Euro NCAP. *Euro NCAP Announces 2026 Protocol Changes to Tackle Modern Driving Risks*. 2026. URL: <https://www.euroncap.com/press-media/euro-ncap-announces-2026-protocol-changes-to-tackle-modern-driving-risks/> (visited on 08/27/2026).
- [7] Luise C. Schaefer et al. "Odds Ratios for Reclined Seating Positions in Real-World Crashes". In: *Accident Analysis & Prevention* 161 (2021), p. 106357. ISSN: 0001-4575. DOI: 10.1016/j.aap.2021.106357. URL: <https://www.sciencedirect.com/science/article/pii/S0001457521003882>.
- [8] National Highway Traffic Safety Administration. *Federal Motor Vehicle Safety Standards; Occupant Crash Protection. Final Rule*. 65 Federal Register 30680, Docket No. NHTSA 00-7013. Advanced air bag rule. 2000.
- [9] National Highway Traffic Safety Administration. *Air Bags*. 2026. URL: <https://www.nhtsa.gov/vehicle-safety/air-bags> (visited on 08/27/2026).
- [10] Augustus B. Chidester, Eric Ferguson, and John C. Kindelberger. "Air Bag Crash Investigations". In: (). National Highway Traffic Safety Administration, Paper No. 299.
- [11] Luis A. Salazar-Calderon, Sergio Alberto Navarro-Tuch, and Javier Izquierdo-Reyes. "Framework, Implementation, and User Experience Aspects of Driver Monitoring: A Systematic Review". In: *Applied Sciences* 15.21 (2025), p. 11638. DOI: 10.3390/app152111638. URL: <https://www.mdpi.com/2076-3417/15/21/11638>.

- [12] Matthew P. Reed, Sheila M. Ebert, and Lawrence W. Schneider. “Development of Performance Specifications for the Occupant Classification Anthropomorphic Test Device (OCATD)”. In: (2001). Paper Number 411.
- [13] Juan Diego Ortega et al. “Challenges of Large-Scale Multi-Camera Datasets for Driver Monitoring Systems”. In: *Sensors* 22.7 (2022), p. 2554. DOI: 10.3390/s22072554. URL: <https://www.mdpi.com/1424-8220/22/7/2554>.
- [14] Mingming Zhao et al. “Exploration of Driver Posture Monitoring Using Pressure Sensors with Lower Resolution”. In: *Sensors* 21.10 (2021). ISSN: 1424-8220. DOI: 10.3390/s21103346. URL: <https://www.mdpi.com/1424-8220/21/10/3346>.
- [15] Boren Wang et al. “Deep Learning-Based Sitting Posture Recognition from Pressure Distribution Across Hard and Soft Seat Environment”. In: *Applied Sciences* 15.23 (2025), p. 12744. ISSN: 2076-3417. DOI: 10.3390/app152312744. URL: <https://www.mdpi.com/2076-3417/15/23/12744>.
- [16] Qinxiao Quan et al. “Multi-Modal Fusion in Ergonomic Health: Bridging Visual and Pressure for Sitting Posture Detection”. In: *CCF Transactions on Pervasive Computing and Interaction* 6.4 (2024), pp. 380–393. ISSN: 2524-521X, 2524-5228. DOI: 10.1007/s42486-024-00164-x. URL: <https://link.springer.com/10.1007/s42486-024-00164-x>.
- [17] Takaaki Teshima, Masahiro Niitsuma, and Hidekazu Nishimura. “Determining the Onset of Driver’s Preparatory Action for Take-Over in Automated Driving Using Multimodal Data”. In: *Expert Systems with Applications* 246 (2024), p. 123153. ISSN: 0957-4174. DOI: 10.1016/j.eswa.2024.123153. URL: <https://linkinghub.elsevier.com/retrieve/pii/S0957417424000186>.
- [18] Hongfei Xue et al. “DeepFusion: A Deep Learning Framework for the Fusion of Heterogeneous Sensory Data”. In: *Proceedings of the Twentieth ACM International Symposium on Mobile Ad Hoc Networking and Computing. MobiHoc ’19*. Catania, Italy: Association for Computing Machinery, 2019, pp. 151–160. DOI: 10.1145/3323679.3326513. URL: <https://doi.org/10.1145/3323679.3326513>.
- [19] Euro NCAP. *Euro NCAP Assessment Protocol – Occupant Status Monitoring, Version 1.2*. Tech. rep. ONM-2023-1.2. Leuven, Belgium: European New Car Assessment Programme, 2023.
- [20] Anna Müller, Thomas Becker, and Sebastian Hoffmann. “Seat Occupancy Classification Using Capacitive Pressure Mats in Passenger Vehicles”. In: *IEEE Transactions on Intelligent Transportation Systems* 20.8 (2019), pp. 2984–2993. DOI: 10.1109/TITS.2019.2903841.
- [21] Mingming Zhao et al. “Exploration of Driver Posture Monitoring Using Pressure Sensors with Lower Resolution”. In: *Sensors* 21.10 (2021). ISSN: 1424-8220. DOI: 10.3390/s21103346. URL: <https://www.mdpi.com/1424-8220/21/10/3346>.

- [22] Julia Schmidt, Felix Wagner, and Markus Braun. “Driver and Passenger Monitoring via In-Cabin Infrared Cameras: A Real-Time Pose Estimation Approach”. In: *Proceedings of the IEEE Intelligent Vehicles Symposium (IV)*. Las Vegas, NV, USA, 2020, pp. 1122–1129. DOI: 10.1109/IV47632.2020.9304721.
- [23] Wei Chen, Xin Li, and Yan Zhou. “Multimodal Deep Fusion Networks for Robust Perception: A Survey and Taxonomy”. In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 42.11 (2020), pp. 2745–2761. DOI: 10.1109/TPAMI.2020.2967271.
- [24] Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016. ISBN: 978-0262035613.
- [25] Bruno Oliveira, Marina Costa, and Rui Almeida. “Cross-Modal Attention Fusion for Camera and Depth Sensor Integration in Human Activity Recognition”. In: *Proceedings of the European Conference on Computer Vision (ECCV)*. Tel Aviv, Israel, 2022, pp. 411–427. DOI: 10.1007/978-3-031-19812-0_24.
- [26] Dmitri Ivanov, Sergei Petrov, and Alexei Volkov. “Synthetic Human Body Simulation for Training Pressure-Based Occupant Detection Models”. In: *Proceedings of the International Conference on Robotics and Automation (ICRA)*. Xi’an, China, 2021, pp. 5531–5538. DOI: 10.1109/ICRA48506.2021.9561234.
- [27] Hiroshi Nakamura, Yuki Tanaka, and Kenji Sato. “Weight Estimation from Distributed Seat Pressure Sensor Arrays Using Regression Neural Networks”. In: *Sensors* 21.14 (2021), pp. 4762–4775. DOI: 10.3390/s21144762.
- [28] Laura Fernández, Pablo García, and Miguel Ruiz. “A Review of Pressure-Sensing Approaches for Automotive Occupant Classification Systems”. In: *IEEE Sensors Journal* 22.5 (2022), pp. 3891–3904. DOI: 10.1109/JSEN.2022.3149217.
- [29] Aarav Patel, Nisha Sharma, and Rohan Gupta. “OccupantNet: Lightweight Convolutional Networks for In-Vehicle Occupant Pose Classification”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*. Virtual, 2021, pp. 2765–2774. DOI: 10.1109/CVPRW53098.2021.00312.
- [30] Piotr Kowalski, Ewa Nowak, and Adam Zielinski. “RGB-D Fusion for Robust In-Cabin Body Pose Estimation Under Variable Illumination”. In: *Proceedings of the International Conference on Automotive User Interfaces and Interactive Vehicular Applications (AutomotiveUI)*. Ingolstadt, Germany, 2023, pp. 88–97. DOI: 10.1145/3580585.3606845.
- [31] Erik Lindqvist, Freya Karlsson, and Olof Andersson. “Bridging the Domain Gap: A Sim2Real Study for Synthetic Pressure Heatmap Generation in Vehicle Seats”. In: *Journal of Intelligent Transportation Systems* 27.3 (2023), pp. 301–318. DOI: 10.1080/15472450.2023.2178456.
- [32] Henry M. Clever et al. *Bodies at Rest: 3D Human Pose and Shape Estimation from a Pressure Image using Synthetic Data*. 2020. arXiv: 2004.01166 [cs.CV]. URL: <https://arxiv.org/abs/2004.01166>.

- [33] Hongfei Xue et al. “DeepFusion: A Deep Learning Framework for the Fusion of Heterogeneous Sensory Data”. In: *Proceedings of the Twentieth ACM International Symposium on Mobile Ad Hoc Networking and Computing*. Mobihoc '19. Catania, Italy: Association for Computing Machinery, 2019, 151–160. ISBN: 9781450367646. DOI: 10.1145/3323679.3326513. URL: <https://doi.org/10.1145/3323679.3326513>.

A Statement of Ethics / Consent Form Template

This appendix reproduces the informed consent form template used for all human subject data collection sessions described in Section 5.1. Each participant received and signed a copy of this form prior to any camera or pressure sensor recording.

Participant Information and Consent Form

Study title: Sensor Fusion of Camera and Pressure Sensor Data for Cabin Monitoring System

Institution: Deggendorf Institute of Technology, Campus Cham

Researcher: [CHETHAN SRINIVASAREDDY]

I confirm that I have read and understood the information provided about this study, that I have had the opportunity to ask questions, and that my participation is voluntary and may be withdrawn at any time without giving a reason. I consent to the recording of camera images and pressure sensor data during the described seating configurations, and I understand that this data will be processed in accordance with the General Data Protection Regulation (GDPR) and used solely for the purposes of this research.

Participant name: _____ Date: _____

Signature: _____

B Full Dataset Statistics Table

This appendix provides the extended version of the dataset composition summary introduced in Section 5.6, including per-class sample counts for each split.

Table B.1: Extended per-class dataset statistics by split.

Split	Adult	Child	Child Seat	Object	Unoccupied
Training					
Validation					
Test					
Total					

C Additional Results Figures

This appendix collects additional qualitative result figures that supplement the evaluation presented in Chapter 8.

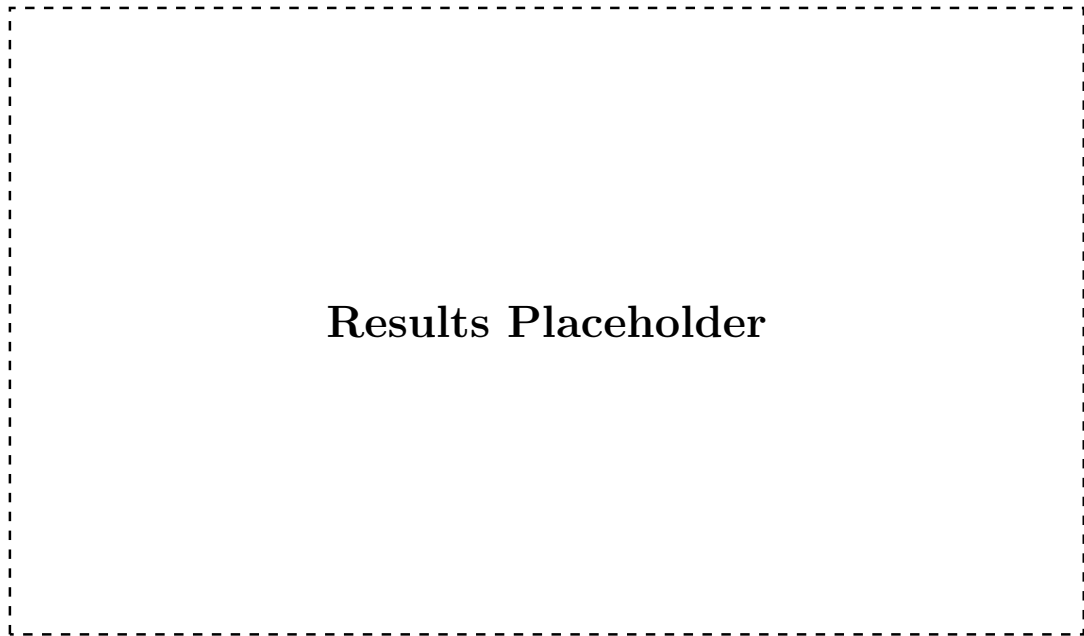


Figure C.1: Additional qualitative results across representative test samples, supplementing Section 8.8.

D Electronic Data Carrier Contents

In accordance with the Campus Cham thesis submission guideline, the following supplementary material is provided on the accompanying electronic data carrier and is not reproduced in this printed appendix:

- Full source code repository, including data preprocessing, synthetic heatmap generation, model training, and evaluation scripts.
- Raw and preprocessed real pressure sensor and camera datasets described in Chapter 5 (subject to the data retention and anonymisation constraints of Section 5.1).
- Trained model checkpoints (weights) for the fused model and all baseline variants evaluated in Chapter 8.
- Configuration files specifying exact training hyperparameters used to reproduce the results reported in this thesis.